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- T^2 Confidence interval
- Bonferroni confidence interval:

* Bonferroni inequality

$\mu_{p\times 1}$.

Given $a \in \mathbb{R}^p$ $\vec{a} \neq 0$

we want CI of: $a^T\mu = \sum a_i \mu_i \in \mathbb{R}$

$\underset{\sim}{x}_1 \cdots \underset{\sim}{x}_n$: Random Sample $\underset{\sim}{X}_{i\, p\times 1}$

$z = a^T \underset{\sim}{X} = a_1x_1 + \cdots + a_px_p$

$Z_1, Z_2, \cdots, Z_n$ is a sample for z from the ~~rand~~ random sample of $\underset{\sim}{X}$

Applying 1-D classical statistics for CI of z.

$(1-\alpha)$ confidence interval

$$a^T\bar{X} \pm \underbrace{\text{Critical value}}_{t_{n-1,\alpha/2}} \sqrt{\frac{s_{zz}}{n}} \qquad \bar{z} = a^T\bar{x}$$

$$s_{zz} = \underset{1\times p}{a^T}\ \underset{p\times p}{S}\ \underset{p\times 1}{a}$$

If pop. is $N_p(\mu, \Sigma)$

& Σ is known

$$a^T\bar{X} \pm \overbrace{\text{Critical value}}^{z_{\alpha/2}} \sqrt{\frac{a^T\Sigma a}{n}}$$

If we drop normal $N_p(\mu,\Sigma)$ condition of population then we have to use S for var($\underset{\sim}{z}$) & we have to use CLT of 1-D on $\underset{\sim}{z}$

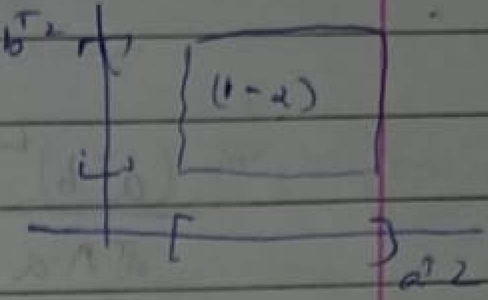
for $n \to \infty$ critical value is $z_{\alpha/2}$

for CI of μ_j, use $a = e_j$.

(c) What is the critical value c for which

$a^T z$

$b^T z$



$$P(a^T\mu \in C_a \ \forall a \in \mathbb{R}^p) = (1-\alpha)$$

$$C_a = \left(a^T\bar{x} \pm c_\alpha\sqrt{\frac{s_{22}}{n}}\right)$$

$$P\left(\bigcap_{\substack{\vec{a}\in\mathbb{R}^p \\ \vec{a}\neq 0}} (a^T\mu \in C_a)\right) = 1-\alpha$$

or what is c s.t. $\forall\ \vec{a}\in\mathbb{R}^p;\ \vec{a}\neq 0$

$$a^T\bar{x} - c\sqrt{\frac{s_{22}}{n}} \le a^T\mu \le a^T\bar{x} + c\sqrt{\frac{s_{22}}{n}}$$

~~For~~ To get this, we use generalized C-S inequality to prove

$$\max_{\substack{\vec{a}\in\mathbb{R}^p \\ \vec{a}\neq 0}} \frac{(\vec{a}^T b)^2}{a^T A a} = \vec{b}^T A^{-1}\vec{b}$$

$\vec{b}\in\mathbb{R}^p$ is fixed & A is pxp given ~~for~~ symm. p.d.

Start with C-S

$$(\vec{a}^T\vec{b})^2 \le (\vec{a}^T\vec{a})(\vec{b}^T\vec{b})$$ & equality holds in (1) iff $\vec{a}$ & $\vec{b}$ are L.D. or $\vec{a} = \lambda\vec{b}$

$$(\vec{a}^T\vec{b})^2 = (\vec{a}A^{1/2}A^{-1/2}\vec{b})^2 \qquad (A = P\Lambda P^T;\ A^{1/2} = P\Lambda^{1/2}P^T)$$

$$= \left((A^{1/2}\vec{a})^T(A^{-1/2}\vec{b})\right)^2$$

$$\le (A^{1/2}\vec{a})^T(A^{1/2}\vec{a})\,(A^{-1/2}\vec{b})^T(A^{-1/2}\vec{b})$$

$$= (\vec{a}A\vec{a})(\vec{b}^T A^{-1}\vec{b})$$

$$\Rightarrow \boxed{(\vec{a}^T\vec{b})^2 \le \underbrace{(a^TAa)}_{>0 \text{ as } A \text{ is p.d.}}(b^TAb)}$$ generalised C.S. Inequality

A is symm-pd

$$\Rightarrow \frac{(a^Tb)^2}{a^TAa} \le b^TA^{-1}b \quad \forall a \in \mathbb{R}^p \;\; a \neq 0$$

equality holds iff $A^{1/2}\vec{a} = \lambda A^{-1/2}\vec{b}$ for $\lambda \in \mathbb{R}, a \neq 0$

or $\vec{a} = \lambda A^{-1}\vec{b}$

upper bound will be attained, ~~at $\vec{a} = A^{-1}b$~~

$$\max_{\substack{\vec{a}\in\mathbb{R}^p \\ \vec{a}\neq 0}} \frac{(a^Tb)^2}{a^TAa} = b^TA^{-1}b$$

$$P\left(\bigcap_{\substack{a\in\mathbb{R}^p \\ a\neq 0}} a^T\mu \in C_a\right) = 1-\alpha$$

$$P\left(\max_{\substack{a\in\mathbb{R}^p \\ a\neq 0}} \left|\frac{a^T\bar{X} - a^T\mu}{\sqrt{\frac{a^TSa}{n}}}\right| \le c\right) = 1-\alpha$$

take $b = \bar{X} - \mu$

$A = S$

$$= P\left(\cancel{\max}\; n(\bar{X}-\mu)^T S^{-1}(\bar{X}-\mu) \le c^2\right) = 1-\alpha$$

$$T^2 = n(\bar{X}-\mu)^T S^{-1}(\bar{X}-\mu)$$

Hotelling T^2 test

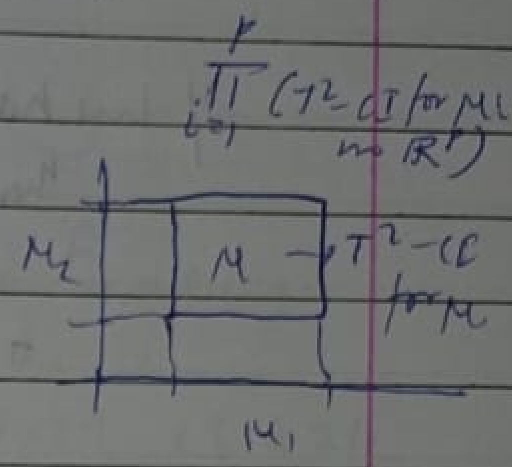
$$a^T\bar{X} \pm c\sqrt{\frac{a^TSa}{n}}$$

c = critical value $= \sqrt{\frac{(n-1)p}{n-p}F_{p,n-p}(\alpha)}$

Largest possible CI for $a^T\mu$ at confidence level $1-\alpha$ $\forall a\in\mathbb{R}^p, a\neq 0$

T^2 - CI for $\mu_i = a^T \mu$ $\qquad a = e_i$

$$= \bar{X}_i \pm c \sqrt{\frac{s_{ii}}{n}} \qquad c = \sqrt{\frac{(n-1)p}{n-p} F_{p,n-p}(\alpha)}$$



T^2-CR is designed $\forall a \in \mathbb{R}^p$ so, it is big enough for CR of $a^T\mu$.
But if we are interested in ~~apso~~ specific vectors $\vec{a}$ only (finite in number) the T^2-CR is big for $a^T\mu$.
We want to ~~sh~~ shrink the CR to take care of CR of ~~a~~ $a^T\mu$ for ~~some~~ specific a's

Bonferroni inequality

$$P(A_1 \cup A_2 \cdots \cup A_k) \le \sum_{i=1}^{k} P(A_i)$$

K: finite
$A_1, \ldots, A_k$ are events

Suppose we are interested in common CR for $a_i^T\mu$ $1 \le i \le k$

$a_i \in \mathbb{R}^p$, $a_i \neq 0$

$$P(a_i^T \mu \in C_i \ \forall\ 1 \le i \le k) = 1-\alpha$$
$$= P\left(\bigcap_{i=1}^{k} a_i^T \mu \in C_i\right)$$
$$\ge 1 - P\left(\bigcup_{i=1}^{k} a_i^T \mu \notin C_i\right)$$
$$\ge 1 - P\left(\bigcup_{i=1}^{k} A_i\right) \underset{(BF)}{\ge} 1 - \sum_{i=1}^{k} P(A_i) = 1 - \sum_{i=1}^{k} P(a_i^T \mu \in C_i) = 1 - \sum_{i=1}^{k} \alpha_i$$

$\therefore 1-\alpha \geq 1-\sum_{i=1}^{k} \alpha_i$

$\Rightarrow \alpha \leq \sum_{i=1}^{k} \alpha_i$

If we have no info on α_i but we have $\alpha \in (0,1)$
then we can assume $\alpha_1 = \alpha_2 = \cdots = \alpha_k = \hat{\alpha}$

$\Rightarrow \frac{\alpha}{k} \leq \hat{\alpha} \Rightarrow \min \hat{\alpha} = \frac{\alpha}{k}$

$\therefore$ BF CI for $a_i^T \mu$ $1 \leq i \leq k$ is given by

$$a_i^T \bar{X} \pm c \sqrt{\frac{a_i^T S a_i}{n}}$$

c = critical value

$= t_{n-1, \left(\frac{\alpha}{2k}\right)}$ $1 \leq i \leq k$

Confidence [redacted] $= \prod_{i=1}^{k}$ CI of $a_i^T \mu$

Let us take $a_i = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} \rightarrow i^{th}$ $k=p$, $1 \leq i \leq p$

Bonferroni CI for μ_i is

$$\bar{X}_i \pm c \sqrt{\frac{s_{ii}}{n}}$$

$c = t_{n-1, \frac{\alpha}{2p}}$

T^2 - CI for μ_i

$$\bar{X}_i \pm c \sqrt{\frac{s_{ii}}{n}}$$

$$c = \sqrt{\frac{(n-1)p}{n-p} F_{p, n-p}(\alpha)}$$

Larger interval

$n = 25 \quad p = 2$

$$\bar{x} = \begin{pmatrix} 4 \\ 5 \end{pmatrix} \quad S = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix}$$

$\alpha = 0.05$

$$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}$$

$(1-\alpha)$ Confidence Ellipsoid for μ: $n(\bar{x}-\mu)^T S^{-1}(\bar{x}-\mu) \le \frac{p(n-1)}{n-p} F_{p,n-p}(\alpha)$

$$25 \begin{pmatrix} 4-\mu_1 & 5-\mu_2 \end{pmatrix} \frac{1}{8}\begin{pmatrix} 3 & -2 \\ -2 & 4 \end{pmatrix}\begin{pmatrix} 4-\mu_1 \\ 5-\mu_2 \end{pmatrix} \le \frac{2\times 24}{23} \times 3.424$$

Simplify to get quadratic inequality in μ_1, μ_2. Centre at $(4, 5)$

T^2-CR for μ_j : $\bar{x}_j \pm (\text{critical value})\sqrt{\frac{s_{jj}}{n}}$

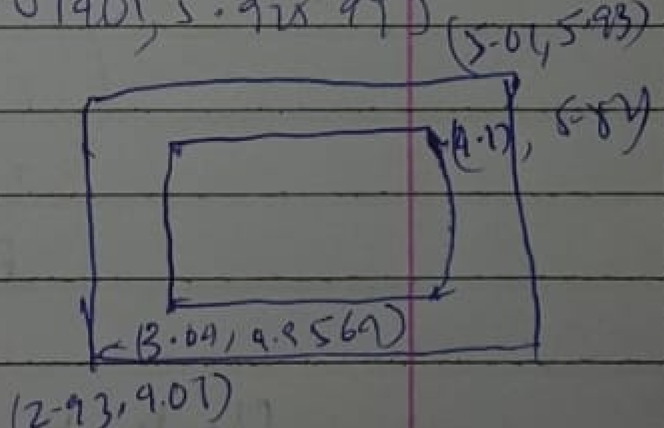
$$c^2 = \sqrt{\frac{p(n-1)}{n-p} F_{p,n-p}(\alpha)}$$

$$\mu_1 : 4 \pm \sqrt{\frac{2\times 24}{23} \times 3.424} \sqrt{\frac{4}{25}}$$

$= [2.93076, 5.06924]$

$$\mu_2 : 5 \pm \sqrt{\frac{2\times 24 \times 3.424}{23}} \sqrt{\frac{3}{25}} = [4.07401, 5.926]$$

T^2-CR for μ: $[2.93076, 5.06924] \times [4.07401, 5.92599]$



Bonferroni CI for μ_j

$$\mu_j = \bar{x}_j \pm \text{critical value} \sqrt{\frac{s_{jj}}{n}}$$

$$\text{critical value} = t_{n-1, \frac{\alpha}{2k}} = t_{24, \frac{0.05}{4}} = 2.391$$

$\mu_1 = 4 \pm 2.391\sqrt{\frac{4}{25}}$ $[3.0436, 4.9564]$

$\mu_2 = 5 \pm 2.391\sqrt{\frac{3}{25}}$ $[4.17, 5.8]$

Paired T-test

Repeated Measure test

(1) Paired T-test

~~Some~~ Same subjects: pre ~~o~~ medication vs. post medication.

T_1 $\boxed{1}$ $\boxed{2}$ ... $\boxed{n}$

T_1 $\boxed{1'}$ $\boxed{2'}$ $\boxed{n'}$

$n = n'$

population: $N_p(\mu_1, \Sigma_1)$ $N_p(\mu_2, \Sigma_2)$ — (or for large n ~~we~~ CLT)

independent (random obs) $\begin{cases} X_{11} & X_{21} \\ X_{12} & X_{22} \\ \vdots & \vdots \\ X_{1n} & X_{2n} \end{cases}$

n: # of subjects / observations / samples

2: treatments / classes

$D_1 = X_{11} - X_{21}$

$D_2 = X_{12} - X_{22}$

$\vdots$

$D_n = X_{1n} - X_{2n}$ $\quad \overline{D} = \frac{1}{n} \sum_{i=1}^{n} D_i = \overline{X_1} - \overline{X_2}$

$H_0 : \mu_1 - \mu_2 = \delta_0$ (or some δ)

$H_1 : \mu_1 - \mu_2 \neq \delta_0$ ~~for some δ~~)

$E(\overline{X}_1 - \overline{X}_2) = E(\overline{X}_1) - E(\overline{X}_2) = \mu_1 - \mu_2$

Under H_0

$(1-\alpha)$ T^2 - Confidence ellipsoid

$$n(\overline{D} - \delta_0)^T S_D^{-1} (\overline{D} - \delta_0) \le \frac{n(p-1)}{n-p} F_{p, n-p}(\alpha)$$

fail to reject H_0 at $(1-\alpha)$ confidence level

Eg. $n=5$ patients

$p = 2$ — BP
— Cholestrol

Medication

Before vs After (is there any ~~difference~~ statistically significant diff in mean)

Subject	BP (before)	(BP after)	Cholestrol (Before)	Cholestrol (After)
1	120	115	200	190
2	130	128	210	205
3	125	120	195	192
4	140	135	220	215
5	135	130	205	200

D_i	BP Before - After	Cholestrol Before - After
D_1	5	10
D_2	2	5
D_3	5	3
D_4	5	5
D_5	5	5

$\bar{D} = \begin{pmatrix} 4.4 \\ 5.6 \end{pmatrix}$

$S_D = \frac{1}{4} \sum_{i=1}^{5} (D_i - \bar{D})(D_i - \bar{D})^T$

$= \begin{pmatrix} 1.3 & 1.3 \\ 1.3 & 5.3 \end{pmatrix}$

~~$S_D^{-1} =$~~

~~$H_0 : \bar{D} = 0$~~

$H_0 : \mu_1 - \mu_2 = 0$

$H_1 : \mu_1 - \mu_2 \neq 0$

$T^2 = n \bar{D}^T S_D^{-1} \bar{D} \cong 81.63$

$\frac{n-p}{p(n-1)} T^2 \cong 30.61$

$F_{p,(n-p)(\alpha)} = F_{2,3,(.05)} = 9.55$

Reject H_0

The sample strongly indicates that the medication ~~contr~~ controls BP & cholestrol effectively (95% confidence)

T^2 - CI

μ_{BP_1} μ_{chol_2}

BP : $\overline{D}_1 \pm$ critical value $\sqrt{\frac{s_{D_1}}{n}} = 4.4 \pm 5.05\sqrt{\frac{1.3}{5}}$

$\parallel$

$\sqrt{\frac{p(n-1)}{n-p} F_{p,(n-p),(\alpha)}}$ $[1.83, 6.97]$

Cholesterol : $5.6 \pm 5.05\sqrt{\frac{5a3}{5}} = [0.9, 10.8]$